ARGUS: acquisition v2 and extended evidence

Project explanation, clock-referenced apparatus implementation, strong reference-design comparisons and claim-support boundaries.

Read this first

The implemented probe captures during excitation and binds a complete clock-described frame before inference. External timing/transfer validation remains necessary. The expanded study retains comparisons against strong baselines and reports synthetic arrival prediction separately from physical localization. Publication and grant are separate procedures, and no numerical acceptance probability is established.

Contents

ARGUS acquisition v2: project explanation and evidence review

ARGUS v2 concurrent reference probe and enabling apparatus

Prospective extended reference-design protocol

ARGUS patent-strengthening assessment and evidence boundary

Source documents and exact commands remain in docs/. Current JSON and JUnit evidence remain in output/release/ and research_results/. The separate manuscript PDF is compiled from paper/main.tex.

Drawings and physical boundary

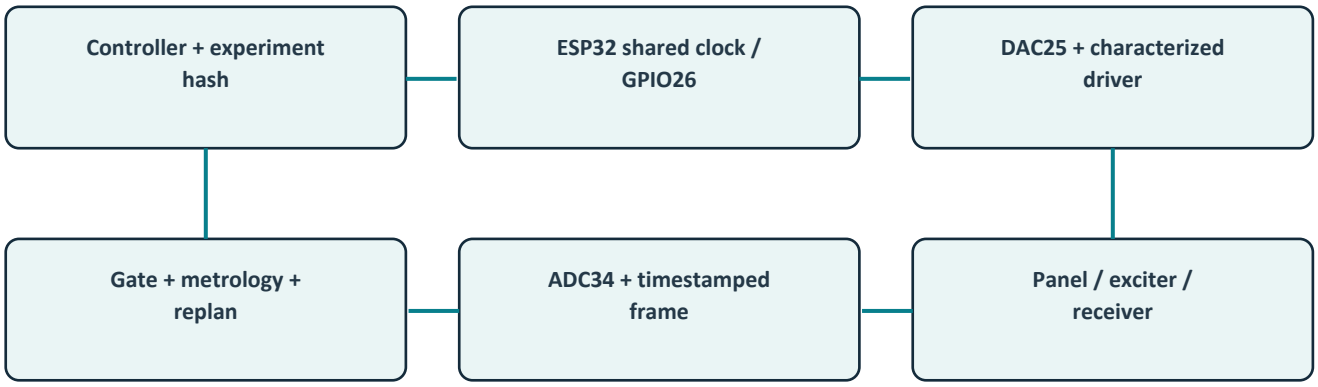


Figure 1. Implemented clock-described v2 acquisition and inference interaction; analog timing and transfer require bench measurements.

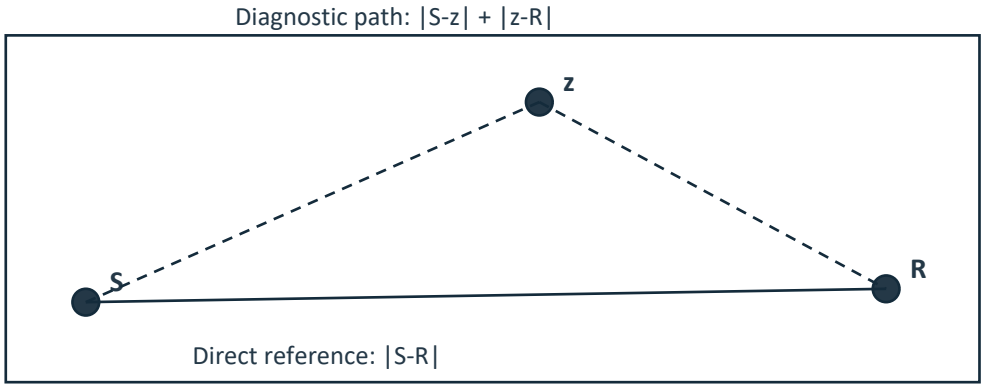


Figure 2. Direct-reference and bistatic diagnostic geometry on a rectangular panel.

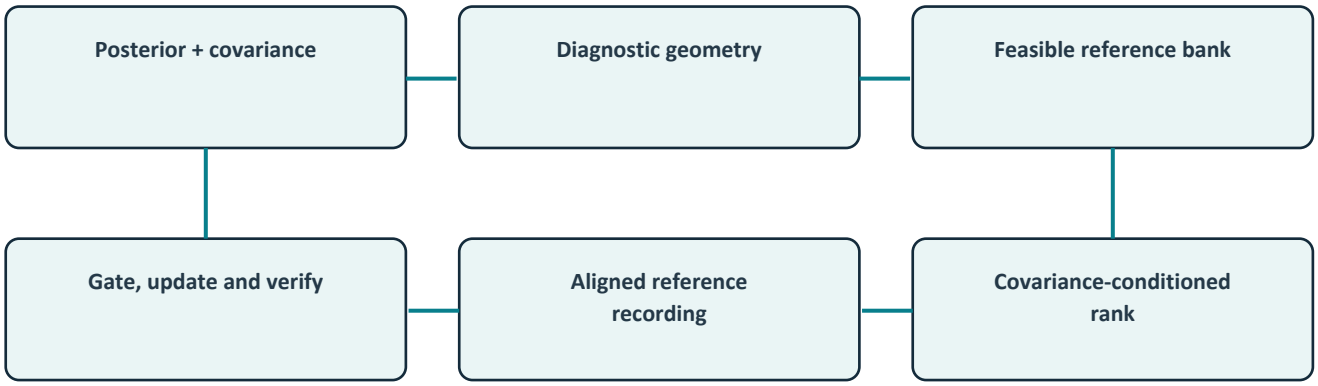


Figure 3. Implemented optional reference selection and gated update sequence. Covariance ranking uses all posterior mass.

ARGUS acquisition v2: project explanation and evidence review

Date: 7 October 2026. This review describes the current apparatus code and software, while keeping older evidence attached to its original source revision. It is a detailed project briefing and a local technical-review package.

The project in one explanation

ARGUS means Adaptive Recursive Guided Uncertainty Sensing. It is a research prototype that chooses the next acoustic experiment on a rectangular panel. Instead of analysing a single fixed recording, it repeats a loop: select a source/receiver position and excitation, acquire the response, check whether the measurement is usable, update a location probability map, and choose another measurement. It can request a propagation reference, a diagnostic measurement, or a fresh verification, and can abstain when the model or measurement is unsupported.

The practical idea is useful when measurement positions and budgets are limited. Different paths provide different constraints on a possible scatterer. The implemented demonstration assumes one dominant scatterer on a known panel. Its map is conditional on that modelling assumption; it does not establish that a defect exists or infer a reliable defect type, depth, size, or severity.

Technology and responsibility of each part

Part	Technology	Responsibility
Operator interface	Next.js, React, TypeScript	Heatmap, signal plots, recommendations, research evidence, acquisition setup and human controls
API and sessions	Python, FastAPI, Pydantic, WebSockets	Bounded requests, synchronized session/device operations, state and event delivery
Scientific computation	NumPy, SciPy	Waveform generation, resampling, matched filtering, probability updates, covariance conditioning and candidate scoring
State and evidence	SQLite, compressed signals, SHA-256 ledger	Persistent history and detection of changed recorded content against a retained checkpoint
Optional models	PyTorch, pickle-free checkpoints	Research surrogate workflow; it is separate from claims of validated live physical inference
Apparatus reference	Classic ESP32, Arduino C++, DAC25, ADC34, GPIO26	Concurrent source/receiver recording with explicit sample times and an external trigger marker

The default server is local to the laptop. An explicit LAN mode uses an operator access key and exact permitted hosts/origins. Hardware execution has per-device locks, session locks, an emergency latch and experiment constraints. These software controls are not an independent physical power cutoff.

Physics behind the observation

A vibration source excites the panel. Elastic motion propagates along the panel; a receiver measures the response. A boundary or anomaly can scatter, attenuate or alter the motion. Real panels can be dispersive, anisotropic and multimodal, and coupling or temperature can change the signal without damage. ARGUS uses a compact path-delay model to study the control loop, rather than solving the elastic wave equation or a high-fidelity Lamb-wave finite-element model.

For a source S, receiver R and location hypothesis z:

```
direct path d = distance(S, R)
scatter path ell(z) = distance(S, z) + distance(z, R)
direct arrival = timing_offset + d / effective_speed
scatter arrival = timing_offset + ell(z) / effective_speed
```

Distances use physical panel dimensions, not the normalized interface coordinates. The simulator combines delayed direct and scattered excitations, attenuation, frequency-sensitive gain and damped ringing. Its illustrative 180 m/s effective speed is a simulation parameter, not a certified material constant.

Matched filtering correlates the recording with a declared reference excitation to obtain a delay-related response. The structural likelihood compares these responses across location hypotheses. This requires a known excitation time origin: an arbitrary microphone file does not reveal when the source began. Resampling can change the computational sample rate but cannot recover missing early arrivals or reverse aliasing in the original recording.

At 16 kHz, nominal sample spacing is 62.5 microseconds. At an illustrative 180 m/s, one interval corresponds to 11.25 mm of propagation distance. This is not a localization-resolution guarantee: geometry, multiple paths, noise, dispersion, bandwidth and clock uncertainty also determine identifiability.

How inference and planning work

The location map is a normalized two-dimensional grid. For accepted structural evidence, the runtime combines the prior with a tempered likelihood and normalizes the result. Tempering limits how aggressively an approximate signal model can concentrate the map. Entropy, local probability mass, MAP position and a physical containment radius describe that map; these are not certified correctness probabilities.

Separate metrology variables represent propagation speed, electronics timing, noise and coupling proxies. Speed and timing retain a correlated 2 by 2 covariance. Other variables remain approximate marginal blocks. The implementation therefore does not claim a fully joint posterior over every structural and instrument variable.

The default controller ranks a finite candidate bank using labelled heuristic information, discrimination, movement, energy, redundancy and trust terms. An optional finite-horizon controller branches on at most two scalar arrival observations and compares calibration and diagnosis under a declared loss. An additional optional fast selector chooses the direct reference most relevant to the nominated next diagnostic geometry. It stays off by default and can be enabled through the session API or `--task-calibration` in the demonstration CLI.

Task-conditioned reference mathematics

A reference measurement observes `y = b + d/v + noise`, where b is timing offset and v is effective speed. One path provides one constraint on two unknowns. Repeated short paths can leave uncertainty in predictions for longer paths even when one parameter becomes precise.

The selector uses every cell of the location grid. It computes a weighted target arrival sensitivity matrix W, predicts the speed/timing covariance after each feasible reference and ranks the resulting diagnostic prediction variance plus cost:

```
g(z) = [-ell(z)/v^2, 1]
W = sum_z probability(z) * transpose(g(z)) * g(z)
h(d) = [-d/v^2, 1]
K = C * transpose(h) / (h*C*transpose(h) + R/q)
C_after = (I-K*h)*C*transpose(I-K*h) + K*(R/q)*transpose(K)
objective = v^2 * trace(W*C_after) + declared_acquisition_cost_m2
```

The covariance update is established Gaussian conditioning in Joseph form. Prediction variance is a standard experimental-design objective. The candidate technical contribution is the specific acoustic apparatus/control interaction, not a new Bayesian principle or a new general optimal-design formula. The objective is path-equivalent propagation uncertainty; it is not predicted defect-position error. Declared cost conversion and assumed quality are research preferences rather than measured monetary loss or actuator energy.

What changed in acquisition v2

The earlier firmware excited the panel and only then began ADC capture. It could miss the early response and lacked a verified excitation-relative time origin. The v2 reference firmware precomputes a quantized chirp or tone burst, captures 16 pretrigger samples, drives DAC25 while reading ADC34, and records each sample's elapsed time and commanded DAC code. GPIO26 marks source onset for an external oscilloscope. Timing deadlines use the declared rate instead of labelling a truncated 62-microsecond interval as exactly 16 kHz.

The bounded frame contains a fresh request identifier, complete experiment hash, original sample count/rate, drive family, exact frequencies/amplitude/window, trigger position, drive interval, ordered ADC samples and timestamps, DAC codes, checksum and completion/overrun status. The host checks all of these before any samples can enter inference. Wrong, incomplete, stale, late, unsupported or checksum-inconsistent frames revoke the connection's acquisition permission.

An operator record binds a specific chip identifier, sampling rate and drive family to an external timing/transfer report digest. Its age and declared error bounds are checked. This is an operator attestation; a JSON record and a hash do not prove that bench measurements occurred. The 90-day age bound and error limits are explicit project policies, not a metrology certification standard.

The classic ESP32 DAC adapter supports chirp and equal-endpoint sine/tone burst, 100 Hz through min(6000 Hz, 0.40 times the source sampling rate), amplitude at most 0.60, windows from 20 to 250 ms, and at most 4096 total samples. Coded waveforms and other simulated families are rejected before excitation. DAC codes describe commanded voltage; they do not measure pressure, amplifier power or panel motion.

Physical evidence gate and persistence

The ingestion service supplies a timing assessment to inference through a server-only argument. A client cannot obtain host-checked status by setting a sensor name or a similarly named metadata flag. Missing timing/adapter support keeps the signal available for plots and history but assigns zero structural evidence and blocks calibration, nuisance updates and verification.

The public WAV upload endpoint additionally accepts a bounded optional `sensor_metadata_json` form field. An exact trigger-aligned declaration binds the complete experiment, source rate, zero trimmed trigger index, external report, operator, date, clock/alignment bounds and validated arrival adapter. It is labelled operator-declared, separately from a checked serial frame. Ordinary browser/phone and WAV recordings without this support remain diagnostic-only recordings.

Original serial ADC values, sample times, DAC codes and declaration metadata are stored in diagnostics before the evidence ledger is appended. Tests demonstrate that changing the retained declaration makes integrity verification fail. Checksums and the ledger detect inconsistent content; they do not authenticate the sensor or prevent a privileged operator from replacing an entire history without an independently retained checkpoint.

What to say in Review 1

"ARGUS chooses the next acoustic experiment from the current location map and instrument uncertainty. It can select a propagation reference before diagnosis. The new probe records during excitation, retains actual timing and drive metadata, and blocks unsupported physical recordings before they influence the map. We have software and synthetic-model evidence; physical localization and patent novelty still require independent validation."

If asked why calibration matters, explain that speed and electronics delay can both shift an arrival. Different reference lengths constrain that ambiguity. If asked why a high confidence value is insufficient, explain that a concentrated map can still be wrong under model mismatch. If asked what is new, identify the specific integrated apparatus/control sequence and acknowledge established matched filtering, Bayesian inference and optimal experimental design.

Running and reproducing the project

```
.venv\Scripts\python.exe run_argus.py
.venv\Scripts\python.exe -m pytest backend\tests -q -o addopts=""
.venv\Scripts\python.exe scripts\run_frontend_release_checks.py
powershell -ExecutionPolicy Bypass -File firmware/esp32/tests/run_host_tests.ps1
.venv\Scripts\python.exe scripts\validate_reference_design_extended.py
```

The interface runs at `http://localhost:5173`. Select a simulation session for a fully runnable demonstration without apparatus. A physical session needs an actual characterized source/receiver setup; importing an invented calibration declaration would undermine the evidence. The serial inventory during this review found no connected serial device. No flashing or physical excitation was performed.

The following dated verification and extended-study results are generated from saved check artifacts. They must be rechecked against source hashes after changes.

Dated verification and extended comparison

The full backend suite passed 470 tests, with 1 intentionally skipped legacy-pickle checkpoint test. A nonfatal framework deprecation warning remains. All 30 frontend tests, TypeScript checking, lint and production build passed.

The extended study used 48 independent seed clusters, four fixed diagnostic geometries, two prespecified noise regimes, five reference policies and three opportunities each. Sealed truth/noise were never supplied to selection. The A/D baselines use dimensionless covariance scales and the same declared cost conversion; cyclic/random are charged the same costs.

All 1,920 attempted runs remain in the means and paired contrasts. 1,909 completed without a rejected opportunity; 11 reference-update opportunities were rejected. The failed runs consume their budget and retain the current state. Source hashes were unchanged during the run and match the current backend.

Regime / reference policy	Arrival RMSE (microseconds)	Latent 95% interval coverage	Completed / attempted
nominal prior noise / cyclic	229.82	93.11%	192 / 192

Regime / reference policy	Arrival RMSE (microseconds)	Latent 95% interval coverage	Completed / attempted
nominal prior noise / random	232.30	94.96%	192 / 192
nominal prior noise / scaled a optimal	200.23	92.13%	192 / 192
nominal prior noise / scaled d optimal	190.85	91.72%	192 / 192
nominal prior noise / task conditioned	189.09	91.83%	192 / 192
double acquisition noise / cyclic	419.26	71.25%	191 / 192
double acquisition noise / random	396.96	69.17%	191 / 192
double acquisition noise / scaled a optimal	363.06	65.86%	189 / 192
double acquisition noise / scaled d optimal	342.63	67.41%	189 / 192
double acquisition noise / task conditioned	343.47	66.19%	189 / 192

Pooled paired advantages are comparator minus task RMSE: positive favours task selection. Intervals resample the 48 seed clusters, preserving within-seed geometry/regime dependence. They are exploratory 95% intervals without multiplicity correction.

Comparator	Mean task advantage (microseconds)	Seed-cluster 95% interval	Interpretation
cyclic	58.26	[35.13, 82.52]	Interval above zero in this model
random	48.35	[26.69, 70.91]	Interval above zero in this model
scaled a optimal	15.37	[4.79, 26.57]	Interval above zero in this model
scaled d optimal	0.46	[-5.70, 6.90]	Inconclusive difference

The D-optimal comparison is inconclusive for both pooled arrival RMSE and prediction-plus-cost. Task RMSE is slightly worse than D-optimal under doubled noise. No general superiority claim is supported. Under nominal noise, task selection improves arrival prediction while worsening timing-offset parameter error (186.23 versus 168.23 microseconds for cyclic). These different outcomes must not be conflated.

Task latent-arrival interval coverage is 91.83% under nominal noise and 66.19% under doubled noise. These first-order parameter intervals do not achieve nominal 95% coverage in the study and do not certify physical

defect confidence. The rejection reason retained by the study is: Direct-path observation is outside propagation-model support (11 opportunities).

The earlier 192-run scalar study and 48-run waveform localization study remain historical, source-bound evidence. This larger scalar study does not replace the need for measured waveforms, independent defect truth or a blind physical campaign.

What remains before stronger publication and patent claims

The code and disclosure can now undergo a substantive technical review. A measured timing/transfer bench and blind specimen-held-out adaptive inspection are still missing. The apparatus compilation status is stated in PROBE_V2_DESIGN.md. Real inventors, ownership, conception records and earlier public disclosures need confirmation. Claim-specific novelty and inventive-step analysis remains open against close active-calibration and acoustic-reference prior art.

No justified 10/10 publication readiness or 6/10 grant probability follows from the implemented changes. Patent application publication is a filing procedure, whereas substantive grant depends on examination. A manuscript publication also depends on the venue and its reviewers. The package is locally prepared; it has not been filed or published.

ARGUS v2 concurrent reference probe and enabling apparatus

7 October 2026. This document separates implemented firmware and host/mock checks from the physical apparatus and metrology still to be built and measured. No board was flashed and no physical waveform was acquired for these changes.

Technical purpose and implemented boundary

The controller's direct and bistatic arrival models require an observation window tied to the excitation onset. The earlier ESP32 sketch generated up to 12 ms of excitation and only then recorded the response. That sequence omitted early arrivals and assigned a different clock origin from the inference model. The earlier host protocol also did not validate declared counts, ordered indices, clock times or excitation endpoints.

The v2 sketch computes the supported excitation before acquisition, collects 16 pretrigger samples, and changes the source DAC during the same timed loop in which it records the receiver ADC. It reports per-sample acquisition-call times and source DAC codes. It returns unsuccessful status on deadline overrun or stop and disables the waveform before transmitting the buffered result. The host can now reject inconsistent or unsupported acquisitions before defect inference. These are implemented control and data-integrity effects; they do not establish physical localization accuracy or patent novelty.

`argus_protocol.h` contains portable parsing, hard bounds, rational sample deadlines, excitation quantization and checksum logic. `argus_probe.ino` applies that logic to classic ESP32 GPIO25 DAC and GPIO34 ADC through Arduino core 2.x APIs. Other ESP32 variants without this DAC/pin layout are outside this reference. An Arduino mock compiles the entire sketch on the host. An isolated official board-toolchain installation was subsequently attempted with Arduino CLI 1.5.1 and stable Espressif core 3.3.12 for `esp32:esp32:esp32`. The CLI ZIP hash was verified, but the core installation did not finish within the bounded attempt. Its two compiler archives alone require over 1 GiB before the chip libraries and core. No board compile command ran and no board binary exists.

The receipt is `output/release/probe-v2-board-build.json`; isolated CLI/downloads remain under `tmp/toolchains/argus-v2`. No global tool installation was changed.

Apparatus drawing specification

Figure A should show the following numbered elements and connections:

1. Rectangular plate under investigation, with measured width and height.

2. Contact exciter at source position S, connected to an external amplifier.

3. Contact piezoelectric receiver at R, connected to a buffered analog front end.

4. Source DAC on classic ESP32 GPIO25, AC-coupled into the amplifier input.

5. Receiver front end, biased near half the ADC range, protected/clamped and band limited, connected to GPIO34. Show a common electrical ground.

6. Common monotonic microsecond counter controlling DAC updates and recording ADC-call timestamps. Show separate arrows to DAC and ADC; the calls are sequential within each sample interval, not simultaneous converter apertures. GPIO26 supplies a logic trigger marker for external timing instrumentation.

7. A STOP input at GPIO27 and an independent actuator-supply cutoff. The GPIO input is polled software; the independent physical cutoff is a proposed circuit.

8. A USB serial link to the laptop controller, structural probability grid, correlated speed/timing covariance, feasible-action bank and observation gate.

9. A measured positioning fixture or manual coordinate marks for S and R.

Figure B should show 16 idle pretrigger samples, a recorded trigger marker, a windowed source burst while ADC acquisition continues, a quiet receiver tail, waveform disable, and buffered frame transmission. Mark the difference between the DAC-call trigger and actual electrical/ADC aperture timing, which must be measured externally.

Figure C should show alternative direct-reference paths of 0.18, 0.36 and 0.54 m, the correlated (velocity, delay) covariance, a chosen diagnostic bistatic path, and the controller selecting the reference that reduces its full-grid propagation prediction variance. Distinct reference lengths give different observation directions; repeating one length cannot independently identify both propagation velocity and instrument delay.

Analog and mechanical embodiment proposed for construction

Use a 600 x 400 mm plate, a fixed coordinate origin, an independently measured source/receiver position and a known healthy reference arrangement. Material, thickness, boundary fixtures, temperature, exciter and receiver part numbers, front-end transfer response, coupling layer and gain must be recorded from the actual apparatus. The default software word `composite` and effective speed 180 m/s are simulator settings, not a physical material specification.

GPIO25 is a low-current 8-bit DAC output. It must drive an AC-coupled, high-impedance amplifier input, never a speaker, motor or low-impedance piezo directly. The reference idle code is 128. For amplitude a, the requested source code is round(128 + 127 a u[n]), with a <= 0.60 and |u[n]| <= 1. The maximum nominal excursion is therefore 76 DAC codes from idle. This bound is on digital drive commands. It is not a measured force, acoustic pressure, supply current, energy, power or transducer displacement bound. Establish external amplifier gain, voltage/current clamps and the permissible excitation from the actual specimen/transducer specification before energizing it.

A piezoelectric receiver requires a high-impedance buffer, input protection, an appropriate antialias filter and bias into the 0-3.3 V ADC range. ESP32 ADC offset, nonlinearity, saturation, clock drift and DAC transfer characteristics must be measured rather than inferred from their bit depths. A direct electrical or healthy-panel transfer reference must justify the host's normalized model units. The affine raw-code conversion (raw - 2048)/2048 alone does not characterize the acoustic transfer function or coupling. A measured receiver reference and operator calibration declaration must be retained separately.

Use a mechanical source/receiver fixture where possible. If manually placed, record the measured coordinates and placement uncertainty for each observation. A camera overlay assists placement but does not certify absolute probe position. No robotic pose or force-controlled coupling is implemented in this embodiment.

Frozen version 2 wire contract

All lines are ASCII, newline terminated. Identity/hash fields are lowercase hexadecimal. Firmware supports `PING`, `CAPABILITIES` (aliases `CAPS`, `STATUS`), `STOP`, `RESET_STOP`, and the v2 experiment command. Legacy `EXPERIMENT`, `EXCITE`, `READ` and unsupported waveform names are rejected before driving.

```
PING
PONG,ARGUS_PROBE,2

CAPABILITIES
CAPS,2,ESP32_DAC25_ADC34_<12lowerhex_chip_id>,16000,4096,12,concurrent_triggered,DAC
  8_SAMPLED_SINE,CHIRP|TONE_BURST,16

EXPERIMENT2 <id32hex> <experiment_sha256> <fstart> <fend> <amplitude> <window_us>
  <CHIRP|TONE_BURST> <requested_rate> <pretrigger_count>

BEGIN2,<id>,<hash>,<total_count>,<requested_rate>,<adc_bits>,<drive_family>,<wavefor
  m>,<trigger_index>,<trigger_time_us>,<fstart>,<fend>,<amplitude>,<window_us>,<burs
  t_us>,<drive_start_us>,<drive_end_us>
DATA2,<index>,<elapsed_us>,<adc_raw>,<drive_code>
END2,<id>,<total_count>,<fnv1a32_hex>,<status>,<maximum_deadline_lateness_us>
```

The full Experiment hash binds geometry and excitation metadata on the host; firmware echoes the supplied hash rather than calculating or authenticating it. The capability device identifier includes the low 48 bits of ESP.getEfuseMac(), allowing the host to bind the operator's bench declaration to a specific chip. It is not an authenticated or unforgeable sensor identity. FNV-1a detects transfer inconsistency; it is not a cryptographic message-authentication or provenance guarantee.

The checksum starts at 2166136261 and processes exact ASCII lines `index,elapsed_us,adc_raw,drive_code\n` with XOR then multiplication by 16777619, modulo 2^32. The host verifies the echoed id, hash and every declared command parameter, exact ordered count and checksum before accepting a frame.

`elapsed_us` is a wrapping-clock difference from the first ADC-call timestamp. It must start at zero and strictly increase. `trigger_time_us` is the trigger-call start offset at the pretrigger boundary, immediately before setting GPIO26 high and writing the source DAC. `drive_start_us` uses that same scheduling marker; `drive_end_us` is the call time at which the DAC returns to idle after its burst. These timestamps describe firmware call scheduling, not the hardware converter aperture, electrical output settling or transducer-force onset. GPIO26 returns low at the burst-end call and when the source is disabled; its external electrical edges support later timing measurements and remain subject to the actual digitalWrite/DAC/ADC call latencies.

Supported excitation and clock constraints

• Requested rate: 1000-16000 Hz. This is a requested rate, not a claimed measured hardware rate. Each sample reports its actual call-time offset.

• Acquisition window after trigger: 20000-250000 microseconds. The total buffer is 16 + round(window_us * requested_rate / 1000000), at most 4096 samples.

• Source band: 100-6000 Hz, end >= start and end <= 0.40 * requested_rate. This conservative digital margin does not replace an analog antialias filter.

• Supported source families: CHIRP and TONE_BURST. The latter requires identical start/end frequencies. Ricker, multisine, coding and spectral notches are explicitly unsupported by this hardware reference.

• Burst sample count: max(16, round(min(10000, 0.22 * window_us) * rate / 1000000)). Windowing uses Tukey alpha 0.45. CHIRP uses linear frequency sweep with cosine initial phase zero and t1=(burst_samples-1)/rate; TONE_BURST uses sine phase.

• Rational deadline offsets floor(index * 1000000 / requested_rate) avoid the old fixed 62 microsecond step masquerading as an exact 16 kHz clock.

• Deadline lateness greater than floor(1000000/(4 * rate)), with minimum 2 us, yields OVERRUN and disables the waveform. No missing samples are invented. At 16 kHz this deadline tolerance is 15 us. A board may legitimately fail at its advertised maximum request rate; the host must then use a measured supported rate instead of accepting the failed capture.

At each tick the sketch writes the prepared DAC code and then starts the ADC read. Source and receiver acquisition overlap throughout the source burst. The sequential per-tick call order is retained in the description. All sine and envelope computations occur before the timing loop. Serial transmission occurs after the source is disabled, so serial baud rate does not determine capture rate. Long frame transmission time still requires a sufficiently bounded host deadline; it cannot be silently truncated into an accepted window.

STOP and the active-low GPIO27 input latch future drive off and are polled during waiting/acquisition. RESET_STOP is refused while the input is active. This polling implementation does not guarantee interrupt latency or protection against CPU failure. A physical actuator-supply interlock remains necessary where that protection is required. The waveform returns to biased DAC idle, which is safe only with the declared AC-coupled amplifier circuit.

Concrete enabling example, not an acquired measurement

For the 600 x 400 mm coordinate system, place a proposed reference source at normalized (0.05, 0.08) and receiver at (0.35, 0.08). These correspond to (30,32) mm and (210,32) mm, a direct reference path of 180 mm. A second reference (0.05,0.50) to (0.65,0.50) has a 360 mm path; a third (0.05,0.92) to (0.95,0.92) has a 540 mm path. Their placements and uncertainty require actual independent measurement.

Command a 1200-3000 Hz CHIRP, amplitude 0.28, window 120000 us, rate 16000 Hz and 16 pretrigger samples. The portable logic requests 1936 total samples, including 1920 after the trigger, and a 160-sample/10000 us source burst. Record actual call timestamps and the quantized source sequence. After a bench-calibrated, complete frame is accepted, align the trigger and normalize the receiver under the host acquisition contract. Obtain the reference delay from the known excitation. Update the correlated speed/timing block using that one scalar observation; use uniform structural likelihood for calibration.

For the purely illustrative simulator values v=180 m/s and t0=0.8 ms, the three direct arrivals are 1.8, 2.8 and 3.8 ms after the declared trigger. These are worked predictions, not measured arrivals. The v2 window includes them; the earlier excite-then-record sketch started acquisition after such early arrivals.

The task-conditioned reference selector then estimates how each candidate would change propagation variance across all structural grid cells for the pending diagnostic geometry. The chosen accepted reference updates covariance and forward-model means; the controller recomputes the subsequent physical diagnostic action. Its eventual posterior update must use the actual diagnostic waveform, with quality/model-support gates, and fresh verification before a confident stop. A reference action alone never validates defect location.

Patent-strengthening acceptance checklist

Acceptance condition	Evidence status
Distinguish single reference rank-one update from identifying two unknowns	Implemented numerical regression; hardware not established
Rank varied reference lengths by their effect on the pending diagnostic's full-grid propagation variance	Implemented software and synthetic study; scope is propagation prediction
Feed accepted reference means/covariance back into later physically feasible action choice	Implemented controller path; actual apparatus campaign pending
Capture source/receiver concurrently under recorded call-time origin	Implemented v2 sketch; host/mock demonstration only
Bind actual declared family/endpoints, timestamps, DAC sequence and complete count to the action	Implemented protocol and host contract; transfer/clock bench pending
Reject missing, reordered, corrupt, stopped or overrun frames before fusion	Host/portable regression; not sensor authenticity
Reference acquisition cannot supply structural defect likelihood	Implemented uniform calibration likelihood
Fresh diverse quality-supported verification controls confident stop	Implemented controller check; empirical coverage pending
Measure trigger-to-DAC/ADC latency and clock drift on the intended board	Pending physical bench with external timing instrument
Measure transfer/antialias response, gain, noise, clipping and coupling	Pending physical bench
Blind specimen/location-held-out comparison at equal costs and budgets	Pending physical campaign
Map narrow apparatus/control limitations against closest prior art	Professional review pending; no novelty asserted

The hardware/calibration portions cannot be labelled experimentally enabled solely because a portable compiler accepts the code. Preserve the actual circuit revision, firmware checksum, board identity record, external timing captures, operator declaration, raw waveforms and source/receiver metrology with the later physical study.

Host verification actually executed

```
powershell -ExecutionPolicy Bypass -File firmware/esp32/tests/run_host_tests.ps1
```

The available g++ compiler successfully compiled both portable protocol tests and the complete `.ino` sketch against `tests/Arduino.h`, with C++17, warnings, pedantic checking and warnings as errors. The portable tests exercise strict numeric parsing, bounds, unsupported-family rejection, rational rate deadlines, source code limits and checksum sensitivity. The sketch mock exercises source drive during receiver acquisition, pretrigger/quiet windows, complete ordered frames/checksum, active STOP latch, missed-deadline failure and micros rollover.

`output/release/probe-v2-host-fixture.txt` is a deterministic mock frame for host-parser integration. It contains generated ADC values and virtual clock times; it is explicitly not a sensor recording. No board compile, flashing, oscilloscope capture, measured vibration or blind specimen inspection occurred.

Prospective extended reference-design protocol

This protocol is written before the frozen extended study. It tests synthetic direct-reference calibration and propagation prediction. It does not establish defect localization accuracy, measured apparatus performance, novelty, or patentability. The earlier 83000-series study remains a separate artifact.

Fixed design

• Forty-eight new independent seed clusters: `970001 + 31*i`, `i=0..47`.

• Four fixed diagnostic geometries on a 0.60 by 0.40 m panel: short corner, opposite corners, crossed interior probes, and same-side probes. Every target uses a nonuniform 20 by 20 probability grid comprising three Gaussian lobes plus a nonzero background. All posterior cells contribute to each outcome.

• Two prespecified acquisition regimes: sealed nominal-prior noise and twice that acquisition standard deviation. Neither regime changes the estimator's assumed-noise contract. Nominal acquisition uses sealed true velocity; it therefore is not an exactly matched Gaussian oracle.

• Five policies: existing cyclic order, seeded random references, scaled A-optimal, scaled D-optimal, and the implemented task-conditioned selector. All use three identical reference opportunities and the same bank of 0.18, 0.36, 0.54 m paths.

• Total final scope: 48 x 4 x 2 x 5 = 1,920 runs and 5,760 opportunities.

• Current sample rate 16 kHz, assumed quality 0.75, cost conversion 0.0001 m² per existing ARGUS acquisition-cost unit, and three-shot budgets are fixed before results. No oracle, seed filtering, tuning, or result-based extensions are included.

Sealing and observation contract

Before running any policy, generate every case's true velocity and timing, initial means and correlated covariance, nonuniform posterior, target geometry, reference bank, starting pose, three Gaussian innovations, and random policy's indices. Retain their canonical SHA-256 seal and the full realization.

The truth draws use the declared joint Gaussian prior, with velocity clipped to 100–330 m/s and timing to 0.05–2.5 ms. Initial means, covariance magnitudes, and correlations vary prospectively. Clipping is a declared departure from an exact Gaussian prior, not a reason to exclude a run.

An acquisition is the nonlinear scalar delay `t_true + d/v_true + sigma_true*z`. True acquisition sigma is computed once from the sealed true speed and declared initial pose/timing noise, divided by sqrt(0.75), then multiplied by the regime's fixed noise multiplier. It remains independent of every changing policy state. The same step's innovation and true sigma are used for all five paired policies.

The implementation's current estimator-based direct-reference noise estimate and quality 0.75 are used for both candidate design and accepted calibration updates. They never receive true velocity or true acquisition sigma. Every trace retains the actual-to-assumed noise ratio so nominal approximation and deliberate noise mismatch are visible. Sealed truth is used only to create measurements and score held-out target predictions.

Each executed reference is charged the existing additional-experiment, movement, energy, and duration cost from `DecisionLossModel`, with the same conversion. Movement is measured from each policy's preceding acquired pose. Failed calibration consumes its opportunity and full actuation cost. A failed selection consumes an opportunity and the declared additional-experiment cost; the state remains unchanged. Failures are retained in all primary summaries. The cyclic and random policies choose without optimizing cost, but they are charged the same actual cost formula and are included in the same objective.

Strong baseline mathematics

Raw covariance trace or determinant of velocity and timing mixes units. A and D baselines therefore transform covariance using scales `(v_ref, D/v_ref)`, where `v_ref` is the sealed initial speed mean and `D` is the panel diagonal. The scales are fixed across the paired policies and never depend on a diagnostic posterior.

Every baseline candidate uses the same correlated first-order reference update and assumed noise/quality as the task selector. A-optimal minimizes `D² trace(C_scaled_after) + acquisition_cost_m²`. D-optimal minimizes `D² sqrt(det(C_scaled_after)) + acquisition_cost_m²`. At zero cost the criteria are standard A and D design rankings. Their cost-bearing proxies have declared m² design units and are not claimed to be squared localization error. They are strong posterior-independent comparator objectives, not hyperparameters tuned to favor any policy.

Outcomes and inference

The first primary outcome is posterior-weighted held-out diagnostic arrival RMSE in microseconds across all 400 target cells. A second primary outcome is `v_ref² * target_arrival_MSE + actual_total_cost_m²`, using the same sealed speed scale across policies. It is a path-equivalent propagation objective; it is not defect-position error or physical risk.

Retain velocity error, timing error, modeled target variance, posterior-weighted 95% latent-arrival interval coverage, actual costs, rejected opportunities, selection and update latency, failure latency, and per-step alternatives. The coverage measure compares true latent arrival to first-order parameter intervals, not future noisy waveform coverage or empirical defect-confidence calibration.

Report task-minus-comparator performance as comparator minus task, so positive advantages favor the task method. Form a seed-level average across geometries and regimes; bootstrap independent seed clusters 4,000 times with fixed bootstrap seed 118701. Retain each seed's actual contrast and separate geometry-by-regime tables, together with regime and pooled contrasts. The intervals are exploratory and unadjusted for multiple comparisons. Negative, null, heterogeneous, and failed-case outcomes remain in the report; no claims require favorable results.

Reproduction and source boundary

```
.venv\Scripts\python.exe -m pytest
  backend\tests\test_extended_reference_validation.py -q -o addopts=""
.venv\Scripts\python.exe scripts\validate_reference_design_extended.py
```

The default output is `research_results/extended_reference_design_2026_10_07.json`. The test suite runs small fixtures without writing final evidence. Hash this protocol, the owned study script and tests, and every

backend/app Python source before and after the final run. A changed source fingerprint invalidates attribution to a frozen version; preserve that output and rerun only after edits stop. A seal or source hash establishes reproducibility, not hardware provenance or institutional review. Expected runtime is well below ten minutes on the current CPU.

ARGUS patent-strengthening assessment and evidence boundary

Prepared: 7 October 2026. **Purpose:** engineering input for a registered patent agent and technical reviewer. No application, external publication, novelty opinion, freedom-to-operate opinion or grant is represented here. Human inventorship, ownership and previous public disclosures remain unknown.

1. What became more defensible

The supported subject is an acoustic inspection controller linked to a specific source/receiver acquisition contract. It maintains a spatial location distribution and a correlated effective-speed/instrument-delay model; selects a direct reference relevant to the pending diagnostic geometry; records a supported excitation and concurrent receiver response; rejects inconsistent frames before interpretation; updates metrology without adding structural evidence; and replans diagnosis with fresh verification controlling termination.

The v2 reference removes a concrete physical inconsistency in the earlier sketch: excitation finished before recording started, so early arrivals were missing while inference assumed an excitation-aligned observation. The new source DAC and receiver ADC operate within one timed loop, with pretrigger samples, recorded trigger/call timestamps, actual DAC codes and complete action-bound frames. The host requires a device-specific external bench declaration independently of protocol support. These changes support an implementable technical sequence and clearer failure behavior. They do not establish that this sequence is new or non-obvious.

The contribution should be presented as the ordered interaction between reference selection, acquisition validity, propagation feedback and later diagnostic control. Broad ownership over acoustic localization, movable transducers, Bayesian calibration, prediction-variance design, hash chaining or common microcontroller acquisition is unsupported. Adding more components or a longer claim is not evidence of inventive step.

Implementation pointers: `backend/app/active_learning/reference_design.py`, `backend/app/inference/nuisance_posterior.py`, `backend/app/services/engine.py`, `backend/app/hardware/protocol_v2.py`, `backend/app/hardware/devices.py`, and `firmware/esp32/argus_probe.ino`. The complete apparatus and protocol are described in PROBE_V2_DESIGN.md.

2. Closest checked sources and limitations of this search

The following are document-specific observations, not a completed worldwide claim chart. The EPO's EP4592907A1 publication and the NASA-hosted original US10126274B2 patent were read directly. For EP2214009A2, the original publication text reproduced by Google Patents was inspected; direct EPO retrieval failed. Its French wording must be checked against the official document before legal claim analysis. No legal-status conclusion is inferred from mirror labels.

Source and inspected scope	Existing technical subject	Consequence for ARGUS
EP4592907A1, claims 1-3 and 5	Bayesian synchronization of a scientific instrument's digital twin; claim 3 includes measured observations, state weighting/pruning, a variance test and active instrument-setting adjustment for following iterations. The microscope limitation is in dependent claim 5.	A distinction based only on a different instrument or active calibration is weak. Compare the precise reference geometry, separate structural/metrology evidence and later diagnostic control sequence.
EP2214009A2, claims 1, 3-5	Reference/current wave responses, decorrelation, source/receiver exploration probabilities, multiplication into a defect-location probability map, normalization and temporal response correction.	Reference subtraction, probability maps, normalized evidence fusion and timing correction are already adjacent art. Compare the actual ordered reference selection and acquisition-gated feedback.
US10126274B2, claims 1, 4, 6-10	Known-state acquisition using a movable transducer, estimated scattered responses, unknown-state comparison, correlation/deconvolution and adaptive combination weights.	Neither movable sensing nor reference/scattered-wave comparison establishes a new concept. ARGUS currently uses a simpler effective bistatic model, not this patent's validated multi-path response bank.

The first summary is supported by the official EPO EP4592907A1 publication, claims on printed pages 38-39. The second is supported by the EP2214009A2 publication text, with the translation/retrieval qualification above. The third is supported by the original US10126274B2 patent hosted by NASA, claims on printed pages 16-18. These references are relevant even without an opinion on enforceable current claims in any particular country.

The engineering inference is that the broad ingredients face substantial prior art. No checked passage was used to establish the novelty of their combination. A professional must review descriptions as well as claims, cited literature, family documents, examination records and other sequential calibration/control work. Absence of the entire ARGUS sequence from this short table is not proof that nobody disclosed it or that combining the teachings would be non-obvious.

3. The reference-selection mathematics is established design

For structural cell z under pending diagnostic e, ARGUS predicts t(z,e)=t0+L(z,e)/v. Its local derivative with respect to propagation parameters (v,t0) is h_z=[-L(z,e)/v^2,1]. Let C be their covariance and let p_z be all normalized spatial cell masses. The software forms:

```
Q(e,p) = sum_z p_z h_z h_z^T
predicted propagation variance = trace(Q C)
```

A reference path r provides a single scalar arrival observation with derivative h_r and declared noise R. Its local Gaussian conditioning gives a covariance C_r. The selector ranks finite feasible references by:

```
J(r | e,p) = v^2 trace(Q(e,p) C_r) + declared acquisition cost
```

This is a locally linear, posterior-weighted prediction-variance criterion: an application of established V-optimal/Bayesian design, with an added configured cost. Prediction-variance minimization over a finite regressor set, including sequential adaptive variants, predates ARGUS. See Wiens and Li, V-optimal designs for heteroscedastic regression (2014). Bayesian prediction-oriented design has still earlier literature, including O'Hagan's Curve Fitting and Optimal Design for Prediction (1978).

Multiplying timing variance by v^2 produces path-equivalent square metres; it does not convert it into proven defect-location loss. The full-grid weights, current diagnostic geometry and speed/delay covariance give this application a definite physical target, but those features do not make the mathematical criterion newly invented. Local linearization, assumed independent reference noise, stationary effective speed and finite candidate geometry remain model assumptions. A single path gives a rank-one precision increment; different path lengths improve observability, an established identification principle.

The synthetic comparison supports propagation-prediction behavior under its declared model. It does not prove patentability, field localization, minimum detectable defect size, measurement compression or that every instrument parameter improves. Preserve the worse timing-parameter result and all runs. The extended study includes stronger standard A- and D-optimal controls. Its pooled arrival-RMSE advantage over D-optimal is inconclusive (0.46 us, exploratory 95% seed-cluster bootstrap interval [-5.70,6.90] us), and modeled interval coverage falls under doubled acquisition noise. A cyclic-reference improvement cannot support a claim of general design superiority. These are synthetic scalar-reference findings, not physical localization. See the retained extended study and its protocol.

4. Narrow technical concepts for a patent agent to test

The following are discussion concepts, not a final independent claim set. Each would still require a search for the full combination and obviousness. The inventive-step analysis must explain why the interaction solves a physical problem beyond predictable use of known components.

Concept A: task-conditioned reference-to-diagnosis control. A panel interrogation apparatus has a controllable acoustic source, receiver, excitation-referenced acquisition clock and processor. The processor maintains a spatial location distribution and correlated speed/delay covariance; nominates a feasible diagnostic source/receiver geometry; evaluates direct-reference geometries through the propagation-variance effect over that diagnostic's weighted spatial support; acquires the selected reference; updates propagation only from accepted reference evidence; and recomputes a later diagnostic physical action. Restrict the model and physical modality to what is disclosed.

Concept B: acquisition-contract gate inside that control sequence. Before the reference or diagnostic waveform can change its respective state, the host checks that the capture overlaps source excitation, includes pretrigger and the declared observation window, reports the accepted waveform/endpoints and actual DAC sequence, and has bounded clock deviations, ordered complete samples and valid integrity. The bench declaration is tied to a specific chip and sampling rate. A failed acquisition cannot be relabelled as valid evidence. The technical interaction is protection of the arrival model's physical input, not a general claim to framing, checksum algorithms or software authentication.

Concept C: verified subsequent termination. Following the updated propagation model and accepted diagnostic fusion, a confident stop requires the latest qualifying observation to pass quality/model support and meaningful geometry/excitation diversity and agree with the preceding location region. Reciprocal paths or amplitude-only changes cannot satisfy that condition in the current reciprocal model. This is an implemented control guard; there is no claim that its heuristic confidence is a field-calibrated probability.

Candidate dependent limitations include varied reference lengths, explicitly retained covariance cross terms, all grid mass in the target, known waveform families, pretrigger/quiet-tail windows, hard drive bounds, feasibility/no-go checks and the separation of reference from structural evidence. Exact pins, 8-bit resolution and 16 pretrigger samples are reference implementation details; claim necessity should be assessed rather than asserted. Evidence hashes may support reproducibility but should not carry the independent inventive concept.

A mere list of known devices operating independently would be insufficient as a technical case. Show, in code traces and then physical records, how the reference changes the propagation prediction and the next executed

diagnostic geometry, and how the acquisition gate prevents an invalid observation from creating that state change. This is the precise technical interaction to chart against the closest art.

5. Enablement and physical evidence levels

Evidence level	Actual status and permitted statement
Executable controller/host parser	Implemented source and regressions; model-state transitions and protocol rejection can be reproduced.
Portable waveform/protocol logic	g++ property tests pass: strict tokens/bounds, rational deadlines, bounded DAC sequence and checksum behavior.
Full sketch with Arduino mocks	Compiles and passes host tests for overlapping drive/capture, pretrigger, quiet tail, complete frames, STOP, overrun and clock rollover. The fixture is generated ADC/virtual-clock data.
Real ESP32 board-target build	Official CLI 1.5.1 was verified and installed in isolated directories; requested stable core 3.3.12 did not finish installing. No board compile command ran or board binary was generated; receipt in output/release/probe-v2-board-build.json.
Assembled apparatus/clock bench	No measured board, amplifier, ADC aperture, DAC settling, trigger-to-drive latency, drift, transfer or acoustic coupling is represented as validated.
Blind adaptive physical localization	Not performed on this apparatus. No field error, safety or calibrated coverage claim follows from synthetic results.

The v2 sketch's trigger time is a scheduling marker before GPIO26/DAC writes; sample times denote ADC-call start. They are not exact electrical converter apertures. An operator's calibration declaration is a recorded attestation and external evidence digest; the host does not independently certify its truth. Chip identity plus FNV and hashes provide consistency/binding, not sensor authenticity or trusted timestamping. GPIO27 STOP is polled software; an independent actuator-supply cutoff remains a proposed circuit.

The reference amplitude limit 0.60 bounds DAC commands around biased idle 128. It is not a measured force, pressure, current, wattage or energy constraint. The AC-coupled external amplifier, input protection, antialias filtering, specimen fixture and transfer characterization must be supplied for the actual apparatus. The effective nondispersive single-scatterer model cannot silently support arbitrary ultrasonic structures, multi-defect reconstruction or depth, type and severity claims. See the enabling probe design.

6. India: technical readiness, application publication and grant differ

India's definition of an invention requires a new product/process, inventive step and industrial applicability. Inventive step requires non-obviousness, alongside the statutory technical-advance/economic-significance language. The worldwide earlier-publication/use boundary matters for novelty. An implemented demo and a better numerical result do not themselves establish these criteria. IP India, section 2.

The statutory exclusions include mathematical methods/algorithms and computer programmes per se, presentation of information, and mere arrangements of known devices working independently. Linking a processor to a transducer does not automatically resolve eligibility or inventive step; a patent agent must assess the claimed technical operation and current practice. IP India, section 3.

A complete specification must describe operation and performance sufficiently, include the applicant's known best method, and have clear claims fairly based on the disclosure. Consequently the actual acquisition, geometry,

calibration and failure handling matter more than broad labels such as AI or digital twin. Physical tests are not asserted here to be a universal legal prerequisite for filing; the current experiment gaps limit the claims and technical performance representations this package can responsibly support. IP India, section 10.

Application publication is disclosure of a filed application. Ordinary secrecy before publication is generally eighteen months from filing or priority, whichever is earlier; an earlier-publication request is provided for. This is separate from substantive examination and grant. Publication does not itself establish that the claims were found patentable. IP India, section 11A, rule 24.

Examination requires a prescribed request. Grant follows the application's being found in order under the Act; it is a separate legal event. A manuscript publication is different again. ARGUS has no filing/publication/grant number created by this work, and should not be called patent pending or patented. IP India, section 11B, section 43.

The applicable procedure and disclosure dates must be reviewed before a filing decision. If relevant parties are resident in India, foreign-filing restrictions also require assessment; residency is not inferred from this project's folder or timezone. IP India, section 39.

7. Exact package still needed for professional review and publication

1. **Human and ownership records:** identify real human contributors for each claimed mechanism; retain dated conception/design notes and contribution statements. Resolve university/project-credit, collaborator, sponsor and competition rights. Do not infer inventorship or conception dates from AI-generated changes or PDF dates.

2. **Disclosure inventory:** capture repository visibility/history, public demos, review slides, reports, uploads, offers and prior filings with exact dates and audiences. Ask the agent to assess the effect in each desired jurisdiction. Do not assume an available grace period or backdate records.

3. **Closest-art chart:** obtain the official French EP2214009A2 copy; chart independent/dependent claims and descriptions from all three references, related families and cited non-patent design/control literature against every proposed limitation. Separate novelty, obviousness and infringement questions; legal status alone cannot substitute for technical comparison.

4. **Best-method apparatus:** retain the final circuit/BOM, board/core/firmware revision, external driver gain/current/voltage limits, sensor protection, antialias response, fixture, specimen dimensions/material/thickness and measured S/R locations with uncertainty. Draw the clock and source/receiver interconnections and separate built from proposed elements.

5. **Timing/transfer bench:** acquire external oscilloscope/logic-analyser captures of GPIO26, DAC output and ADC/known injected reference. Measure call-to-electrical latency, sample jitter, rate drift, clipping, quantization, drive endpoints and transfer response at the actual supported rate. Preserve original records and independently calculate the declared tolerances.

6. **Reference validity:** demonstrate that the accepted reference peak is the intended direct propagation response rather than a stronger echo, electrical pickup or defect scattering. Vary measured path lengths, estimate the joint speed/delay correlation, and retain an independent held-out path. Determine when dispersion, temperature, boundaries or poor coupling require rejection or a different model rather than spurious precision.

7. **Blind physical closure:** prerecord outcomes/budgets; hold out whole specimens or defect locations; retain healthy/reference labels, independent defect truth and every repeat/failure. Compare task references to cyclic and appropriate fixed/random baselines under equal opportunities and cost; include no-reference, repeated-path and acquisition-gate ablations where appropriate. Report arrival prediction, position error, actual observed coverage, false confidence, rejection and stopping behavior separately.

8. **Frozen source/evidence:** fingerprint all relevant application, firmware, controller/configuration and experiment files; preserve raw captures, metadata, command frames, outcomes and source manifests. A later software change invalidates attribution of earlier results to the changed source unless it is explicitly assessed. Preserve negative results and model likelihood/planning differences in the manuscript.

The immediate defensible deliverable is this reproducible technical disclosure and its implementation evidence. Filing scope, professional claim drafting, ownership and publication sequencing require the real records above. No outside patent contact, filing, payment or publication was performed.